

A Technical Introduction to Stacking ensemble

Section 1

Ensemble learning applications In recent years, due to growing computational power, which allows for training in large ensemble learning in a reasonable time frame, the number of ensemble learning applications has grown increasingly. Some of the applications of ensemble classifiers include:

Section 2

Ensemble size While the number of component classifiers of an ensemble has a great impact on the accuracy of prediction, there is a limited number of studies addressing this problem. A priori determining of ensemble size and the volume and velocity of big data streams make this even more crucial for online ensemble classifiers. Mostly statistical tests were used for determining the proper number of components. More recently, a theoretical framework suggested that there is an ideal number of component classifiers for an ensemble such that having more or less than this number of classifiers would deteriorate the accuracy. It is called "the law of diminishing returns in ensemble construction." Their theoretical framework shows that using the same number of independent component classifiers as class labels gives the highest accuracy.

Section 3

Implementations in statistics packages R: Several packages offer Bayesian model averaging tools, including the BMS (an acronym for Bayesian Model Selection) package, the BAS (an acronym for Bayesian Adaptive Sampling) package, and the BMA package. Other packages implementing predictive stacking for Gaussian process models include the spStack package. Python: scikit-learn, a package for machine learning in Python offers packages for ensemble learning including packages for bagging, voting and averaging methods. MATLAB: classification ensembles are implemented in Statistics and Machine Learning Toolbox.

Section 4

Bayesian model combination Bayesian model combination (BMC) is an algorithmic correction to Bayesian model averaging (BMA). Instead of sampling each model in the ensemble individually, it samples from the space of possible ensembles (with model weights drawn randomly from a Dirichlet distribution having uniform parameters). This modification overcomes the tendency of BMA to converge toward giving all the weight to a single model. Although BMC is somewhat more computationally expensive than BMA, it tends to yield dramatically better

results. BMC has been shown to be better on average (with statistical significance) than BMA and bagging. Use of Bayes' law to compute model weights requires computing the probability of the data given each model. Typically, none of the models in the ensemble are exactly the distribution from which the training data were generated, so all of them correctly receive a value close to zero for this term. This would work well if the ensemble were big enough to sample the entire model-space, but this is rarely possible. Consequently, each pattern in the training data will cause the ensemble weight to shift toward the model in the ensemble that is closest to the distribution of the training data. It essentially reduces to an unnecessarily complex method for doing model selection. The possible weightings for an ensemble can be visualized as lying on a simplex. At each vertex of the simplex, all of the weight is given to a single model in the ensemble. BMA converges toward the vertex that is closest to the distribution of the training data. By contrast, BMC converges toward the point where this distribution projects onto the simplex. In other words, instead of selecting the one model that is closest to the generating distribution, it seeks the combination of models that is closest to the generating distribution. The results from BMA can often be approximated by using cross-validation to select the best model from a bucket of models. Likewise, the results from BMC may be approximated by using cross-validation to select the best ensemble combination from a random sampling of possible weightings.

Section 5

Change detection Change detection is an image analysis problem, consisting of the identification of places where the land cover has changed over time. Change detection is widely used in fields such as urban growth, forest and vegetation dynamics, land use and disaster monitoring. The earliest applications of ensemble classifiers in change detection are designed with the majority voting, Bayesian model averaging, and the maximum posterior probability. Given the growth of satellite data over time, the past decade sees more use of time series methods for continuous change detection from image stacks. One example is a Bayesian ensemble changepoint detection method called BEAST, with the software available as a package Rbeast in R, Python, and Matlab.

Section 6

$$e^k = H(p, q^k) - \lambda \sum_{j \neq k} H(q^j, q^k) \quad \{ \text{displaystyle } e^k = H(p, q^k) - \frac{\lambda}{\sum_{j \neq k} H(q^j, q^k)} \}$$

Section 7

Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if

this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more efficient at improving overall accuracy for the same increase in compute, storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well.

By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in anomaly detection.